

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

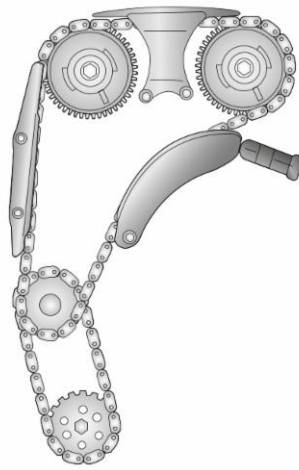
Never turn the engine in the direction opposite to that of normal operation

Note: Avoid bending, torsion or traction

Always use new gaskets

General

Timing chain layout



Removal

Disconnect the battery earth cable

Remove the engine together with the gearbox

Vehicles with manual transmission:

Remove the exhaust pipe support from the gearbox

Remove the engine/gearbox attaching bolts

Separate the gearbox from the engine

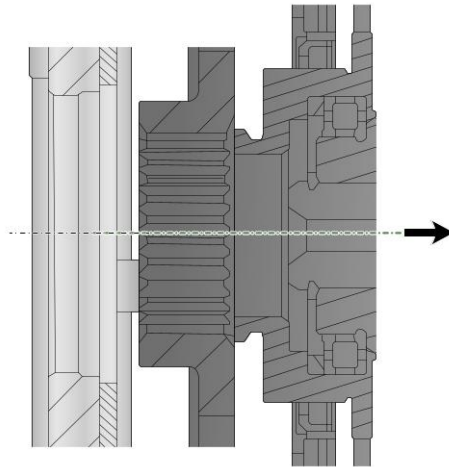
Vehicles with dual-clutch transmission:

Disconnect the gearbox cooling lines

Remove the engine/gearbox attaching bolts

Dual-clutch transmission: Check that the position is correct as shown.

Remove the dual-clutch transmission

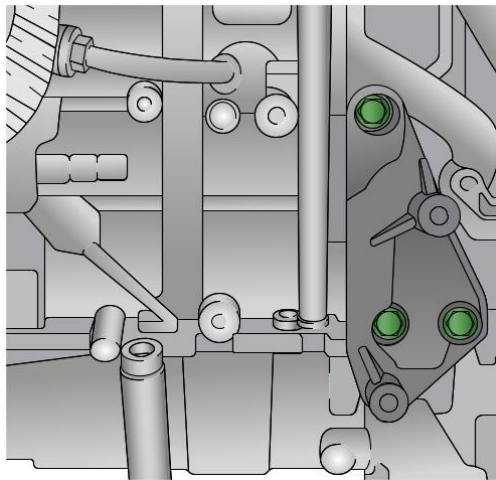
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All vehicles:

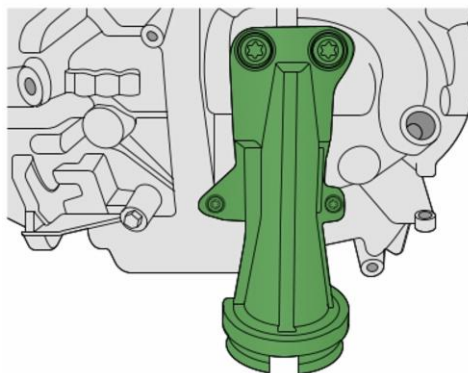
Fit a lifting eye to the cylinder head ((21.5 Nm)) (2 459 012)

Note: Lift the engine (11 7 070, 11 0 020)

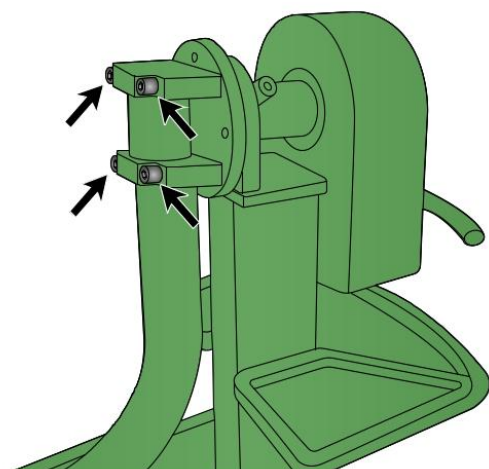
Remove the engine support bracket



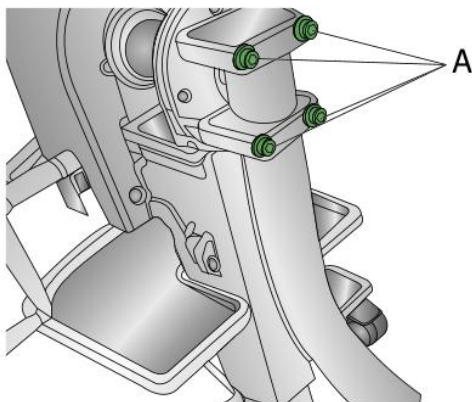
Fit the special tool ((38 Nm)) (2 455 010)



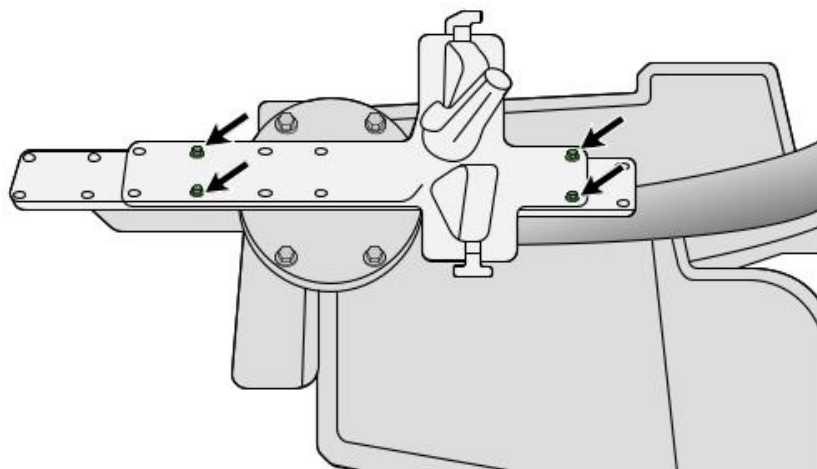
Assemble the special tool as shown (00 2 320, 00 2 300)

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Tighten the special tool bolts ((A: 130 Nm))

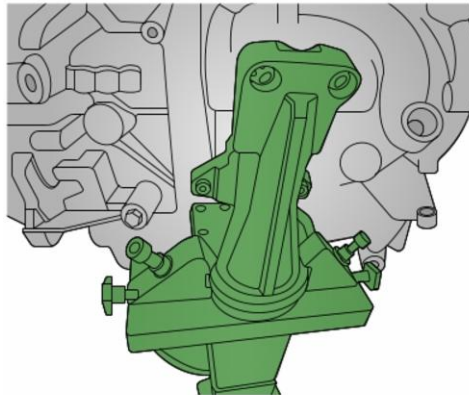


((100 Nm)) (2 413 082, 00 2 320)



Support the engine

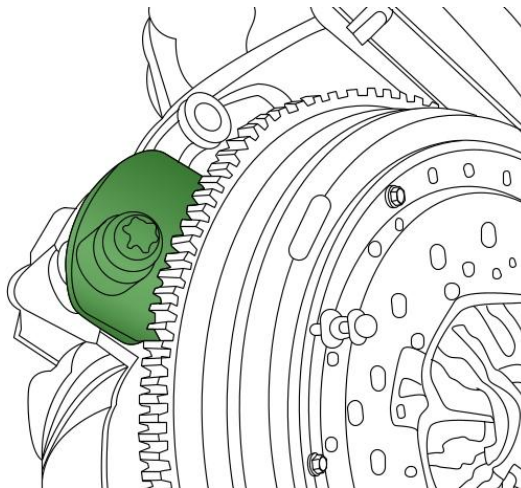
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Remove the cylinder head

Vehicles with manual transmission:

Fit the locking tool (11 9 265, 11 9 811)



Remove the clutch

Remove the flywheel (11 4 180)

Vehicles with dual-clutch transmission:

Fit the locking tool. With washer: ((5 mm)).

Remove the drive plate (11 4 180)

All vehicles:

Remove the crankshaft position sensor gearwheel

Disconnect the oil level sensor

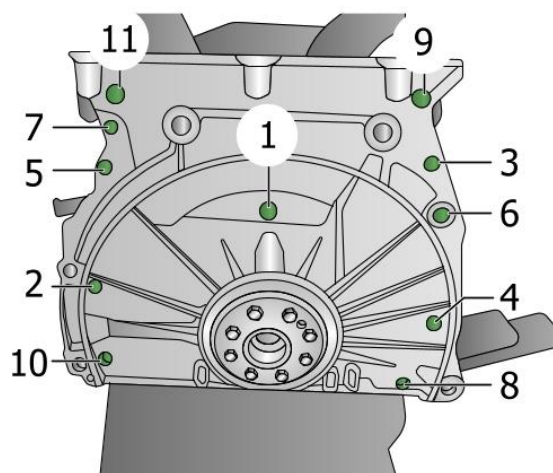
Remove the lower torque reaction rod

Remove the oil sump

Disconnect the crankshaft position sensor

Remove the bolts in the reverse order to that shown ((11 -> 1))

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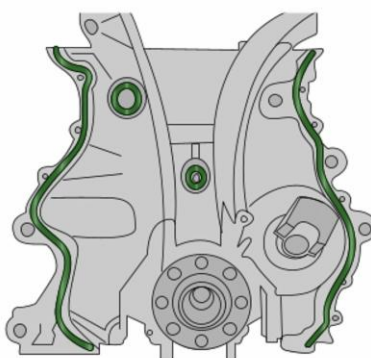
Remove the timing chain cover
 Remove the chain tensioner guide
 Remove the chain guide
 Remove the timing chain
 Remove the oil pump gearwheel
 Remove the oil pump chain

Installation

Note: Clean off the sealing surfaces
 Installation is in the reverse order of removal

Timing chain cover:

Remove and discard the crankshaft oil seal
 Check the dowel pins for damage
 Apply sealant as indicated ((2.0 - 2.5 mm))



Note: The cover must be fitted within 10 minutes of applying the sealant

Refit the timing cover

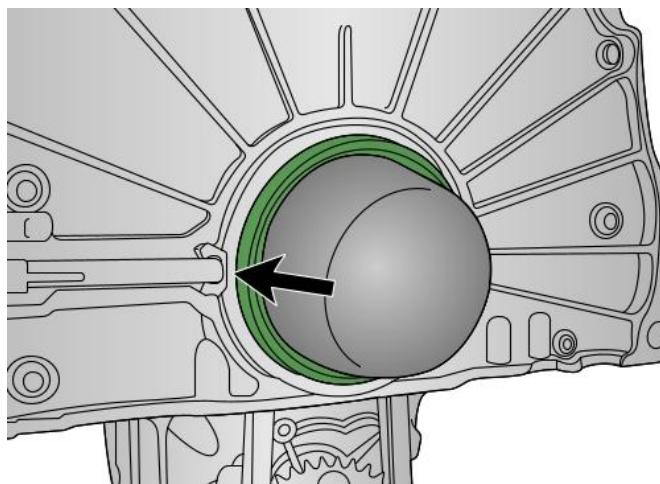
Reconnect the crankshaft position sensor

Fit the special tool (11 8 815)

Note: Do not apply lubricant

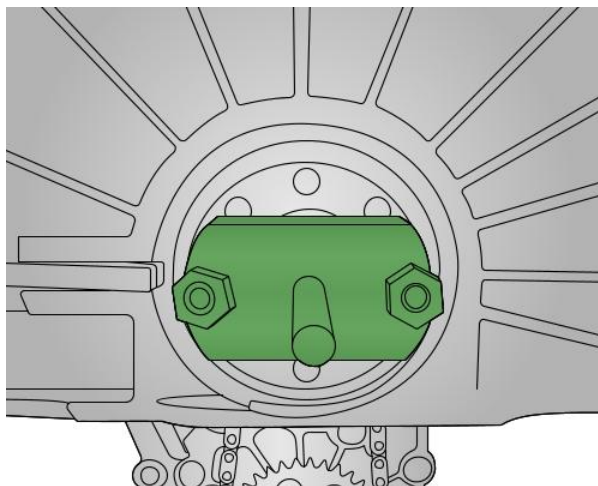
Fit the crankshaft seal over the special tool

Note: Do not touch the lip of the seal

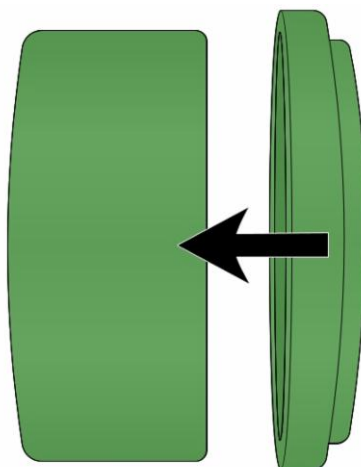
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Remove the special tool (11 8 815)

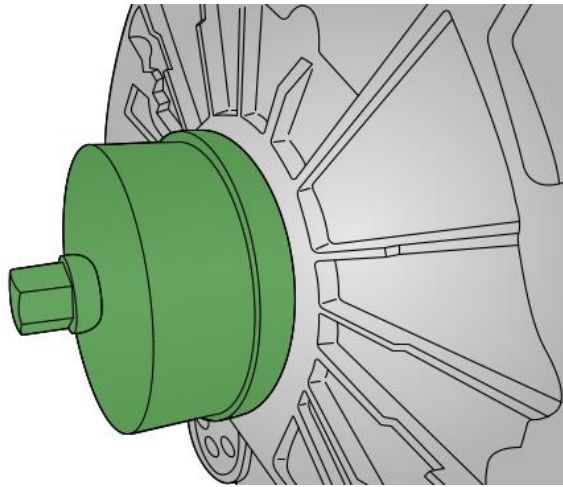
Fit the special tools (11 9 181, 11 9 182, 11 9 183, 11 9 184, 11 9 185)



Assemble the special tool as shown (2 360 215)



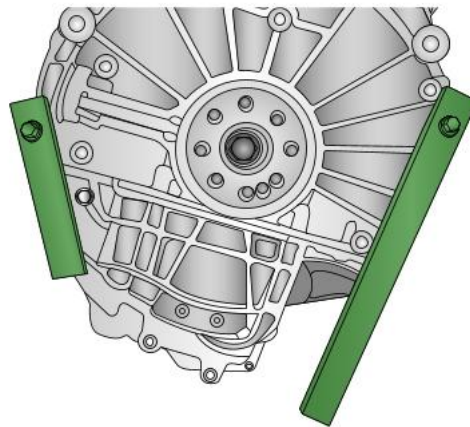
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Tighten the special tool bolt until full surface contact is made

Refit the oil sump

Use the special tools (11 8 780)



Further refitting is in the reverse order of removal

Reconnect the battery earth cable

Torque settings

Oil pump gearwheel

Oil pump gearwheel (Left-hand thread)

Renew the bolt

Stage 1 5 (Nm)

Stage 2 90 (°)

Timing chain tensioner guide

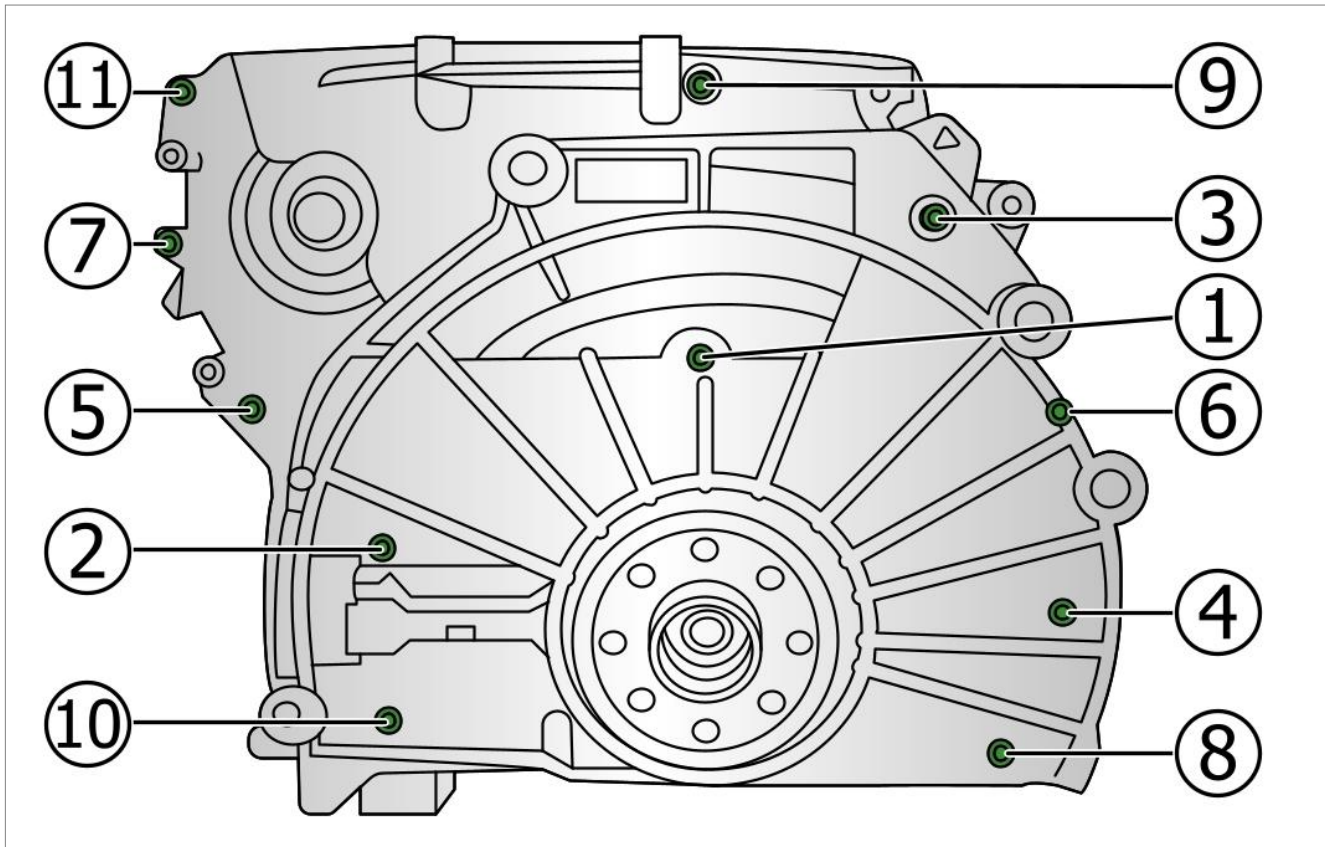
Timing chain tensioner guide (Renew the bolt) 20 (Nm)

Timing chain guide

Timing chain guide (Renew the bolt) 20 (Nm)

Timing cover

Timing cover

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Renew the bolts

Apply sealant 1.5 - 2.0 (mm)

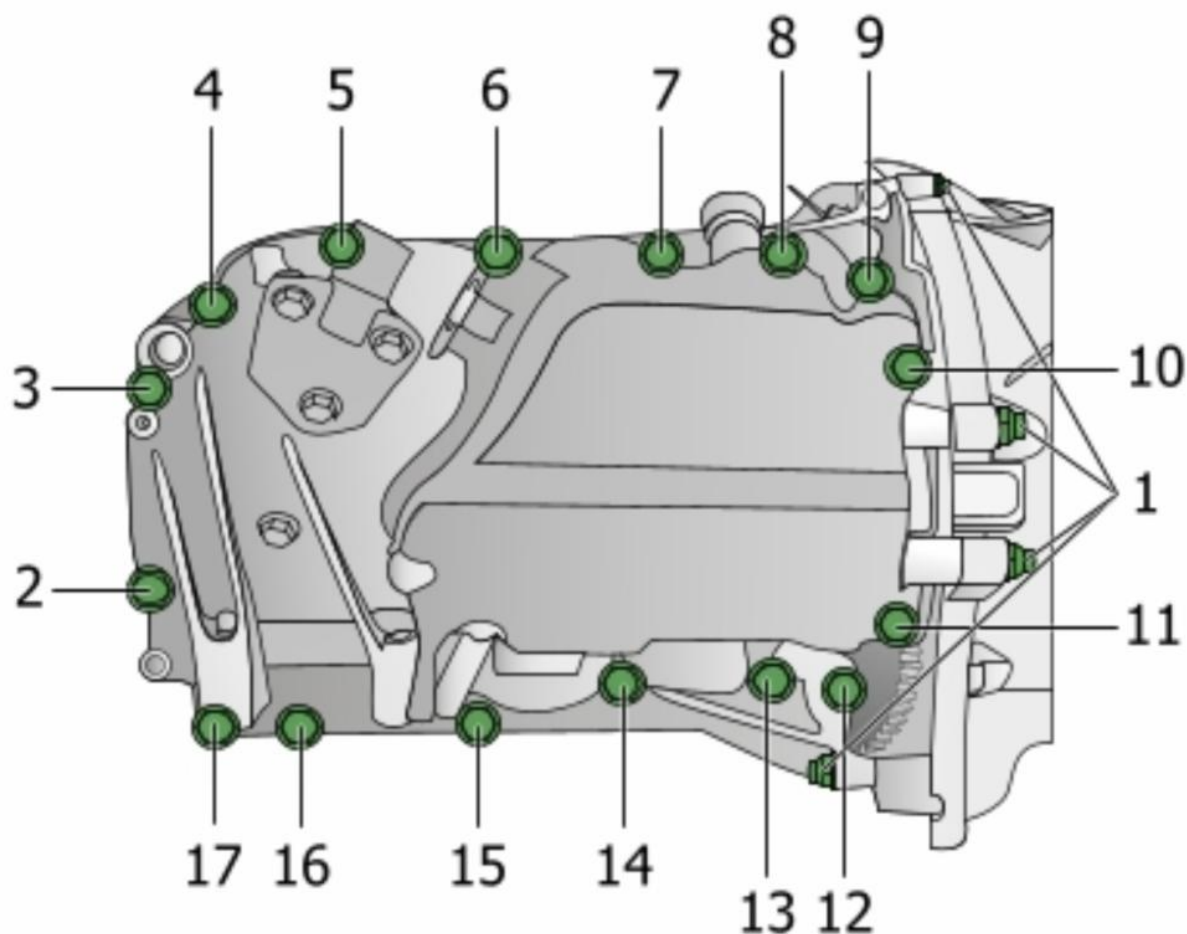
Stage 1 8 (Nm)

Stage 2 90 (°)

Oil sump

Sump (Clean the contact surfaces)

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Apply sealant 2.0 - 2.5 (mm)

Stage 1 (Tighten the bolts until full surface contact is made)

Stage 2 (Bolts 2 - 17) 24 (Nm)

Stage 3 (Bolts 1) 19 (Nm)

Flywheel

Flywheel

Renew the bolts (Tighten the bolts in several stages, in a criss-cross pattern)

Stage 1 45 (Nm)

Stage 2 40 (°)

Engine/gearbox attaching bolts:

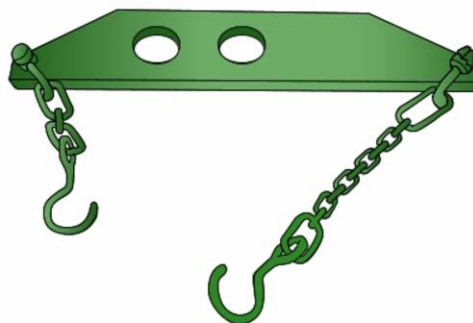
M8: (19 Nm)

M12: (66 Nm)

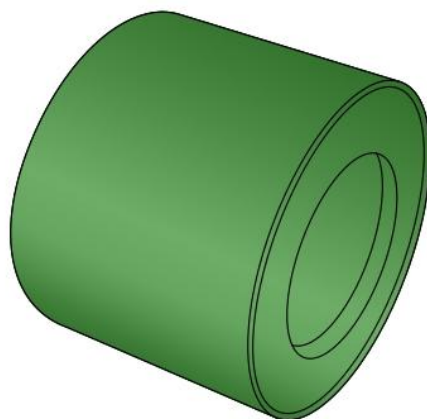
Special tools

Engine support tool: (11 7 070, 11 0 020)

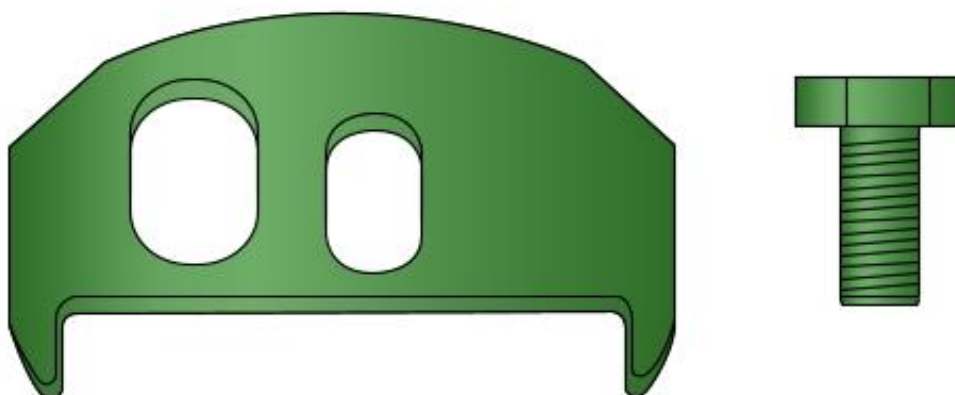
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Bush: (11 9 265)

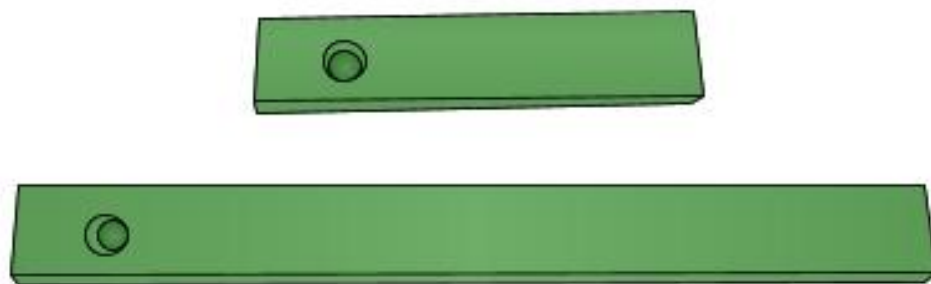


Flywheel locking tool: (11 9 811)

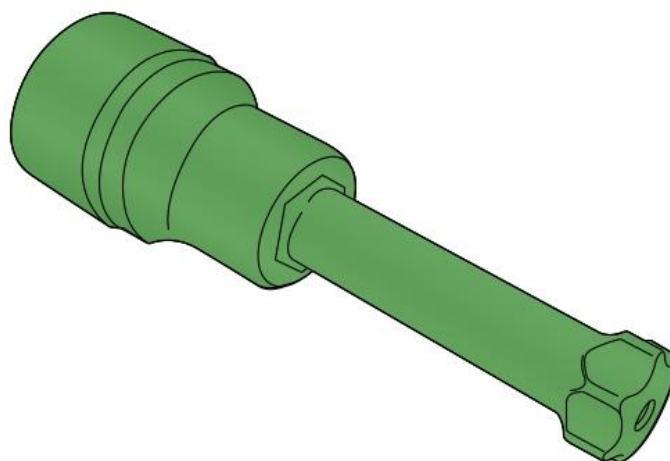


Special tools: (11 8 780)

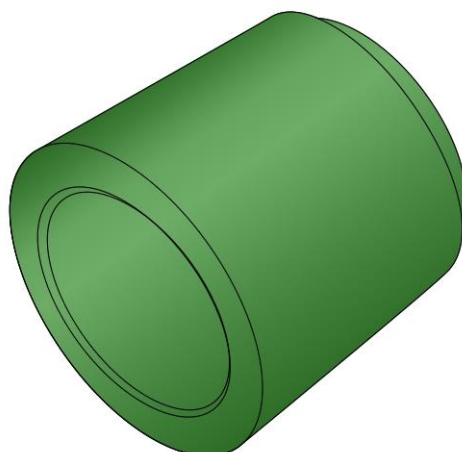
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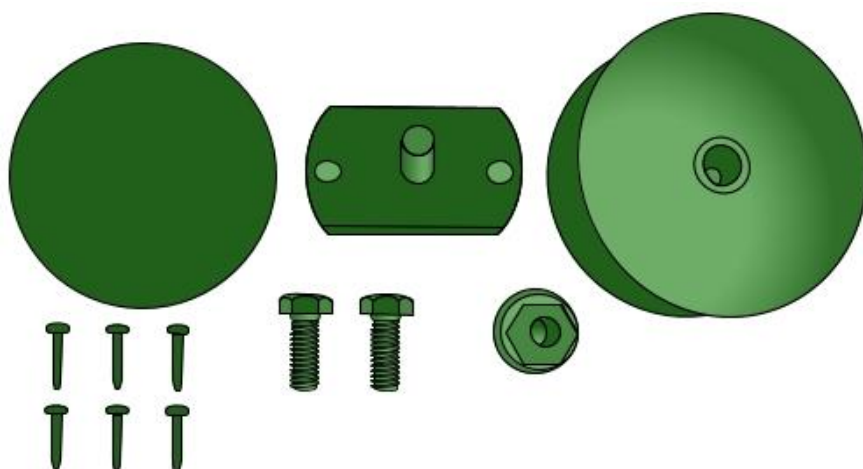
Socket: (11 4 180)



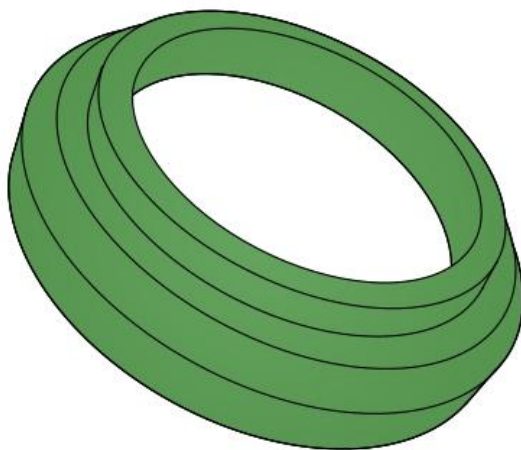
Oil seal fitting tool: (11 8 815)



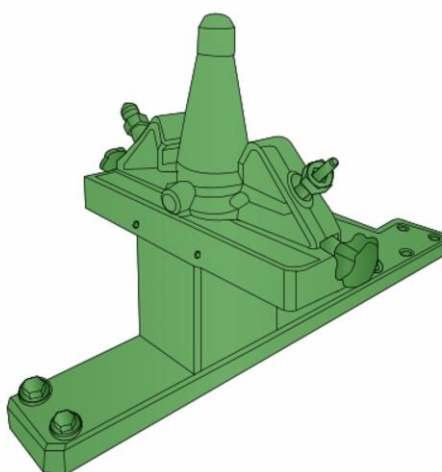
Oil seal fitting tool: (11 9 181, 11 9 182, 11 9 183, 11 9 184, 11 9 185)

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Adaptor: (2 360 215)

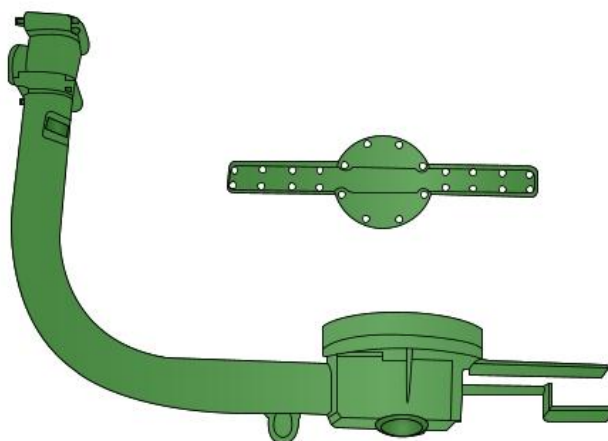


Engine support: (2 413 082)

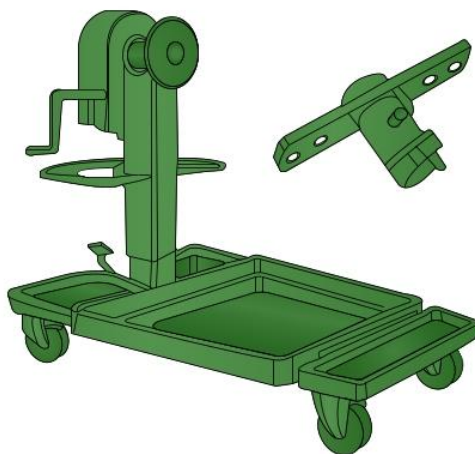


Engine support: (00 2 320)

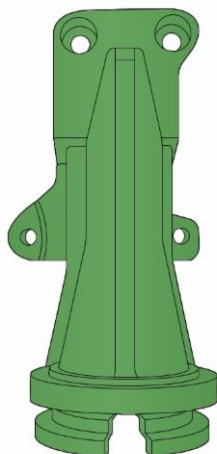
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Engine support: (00 2 300)

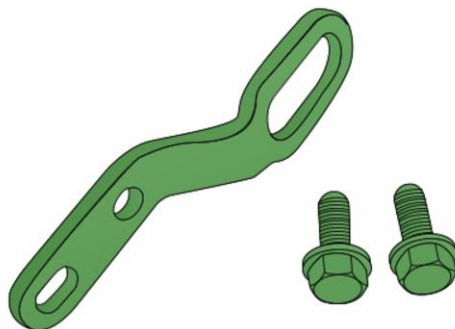


Engine support tool: (2 455 010)



Lifting eye: (2 459 012)

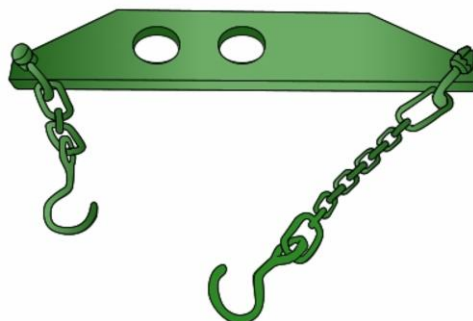
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Socket: (11 6 480)

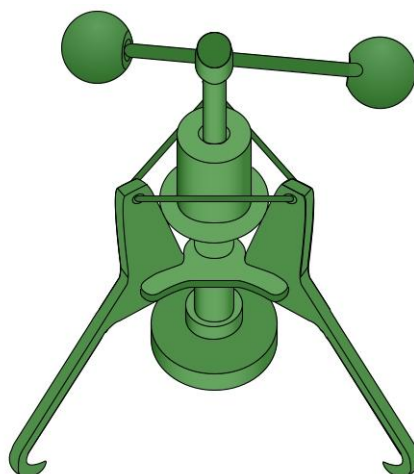


Engine support tool: (11 0 000)

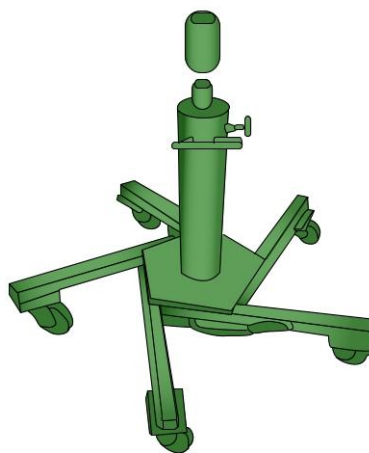


Clutch compressing tool: (21 2 170)

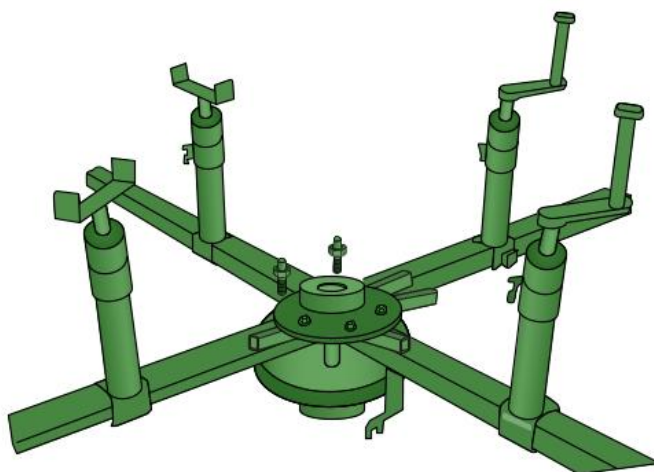
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Transmission jack: (00 2 030)

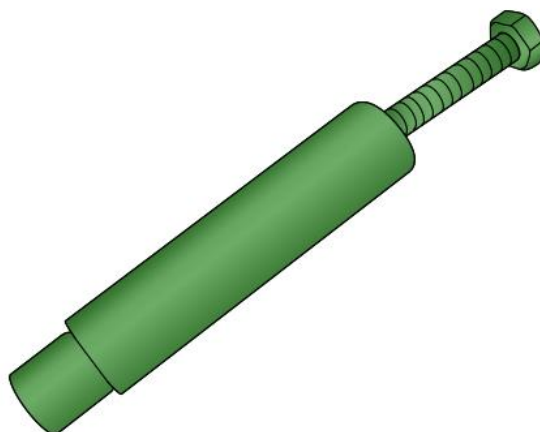


Subframe support: (31 5 250)



Clutch alignment tool: (2 358 471)

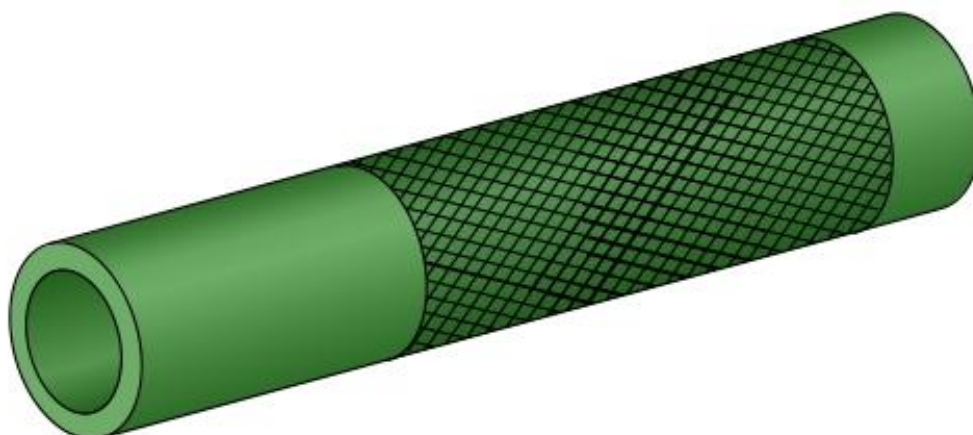
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Flexplate locking tool: (2 288 380)

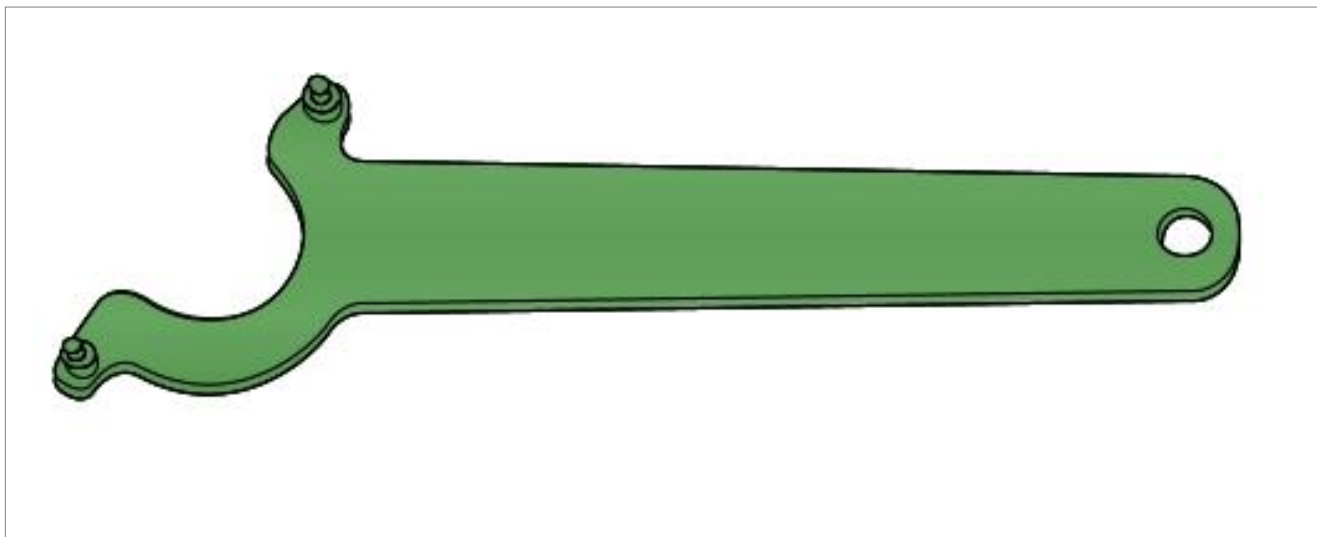


Spark plug removal tool: (12 1 220)

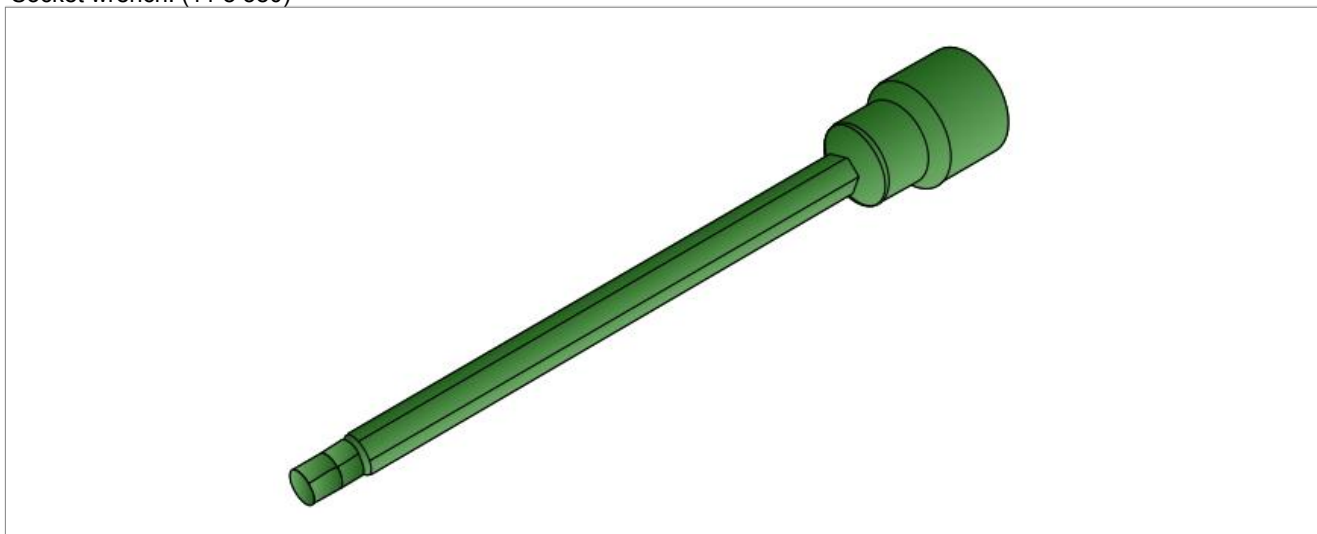


Special tool: (2 360 895)

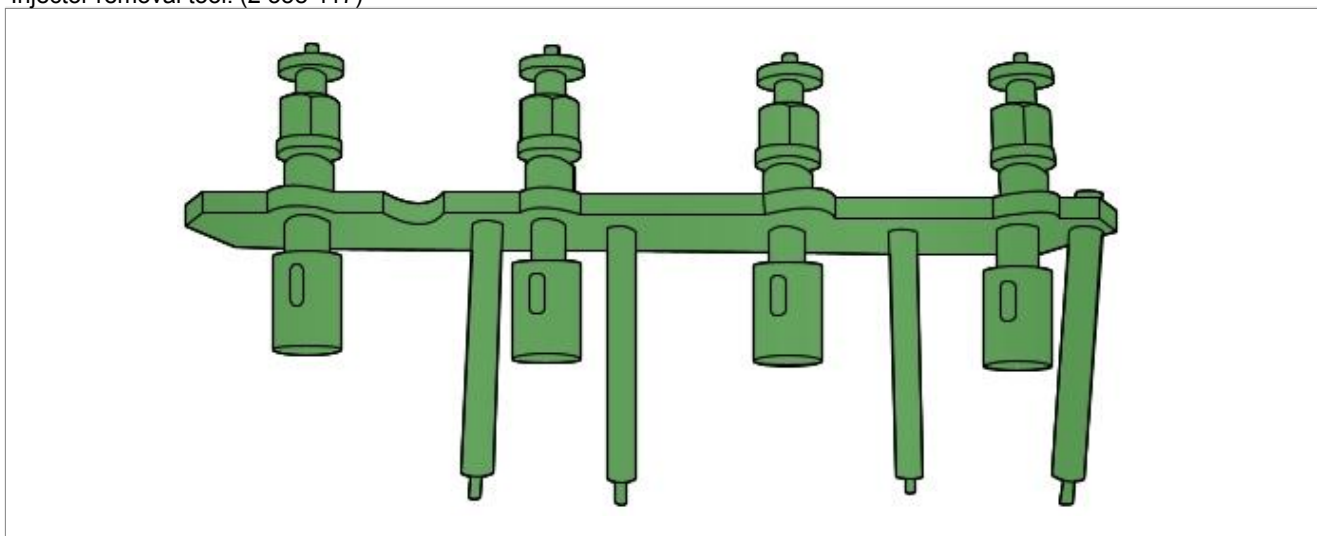
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Socket wrench: (11 8 580)

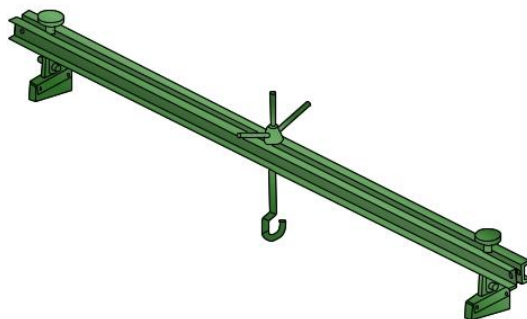


Injector removal tool: (2 358 417)

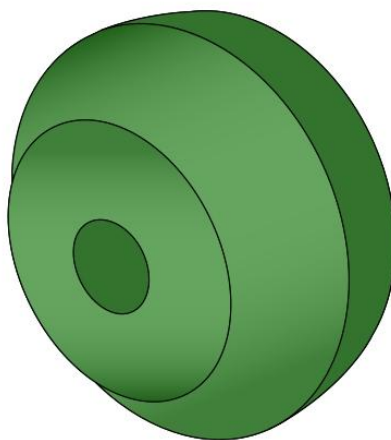


Engine support: (2 414 691)

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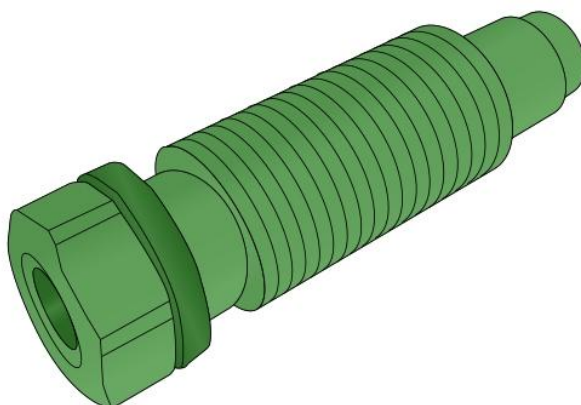
Engine support: (2 357 071)



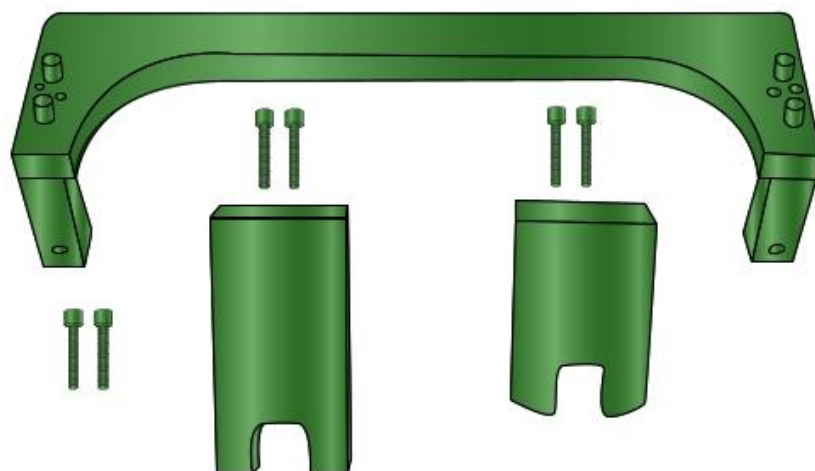
Subframe support: (2 184 136)



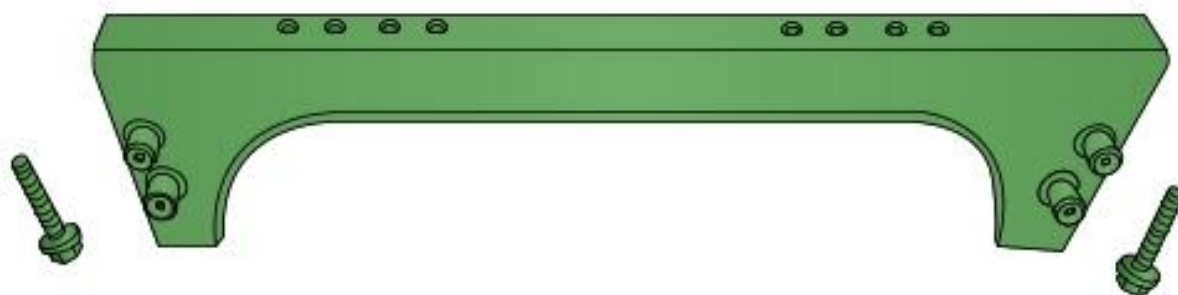
Dummy chain tensioner: (2 455 654)

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Camshaft locking tool: (2 456 372)

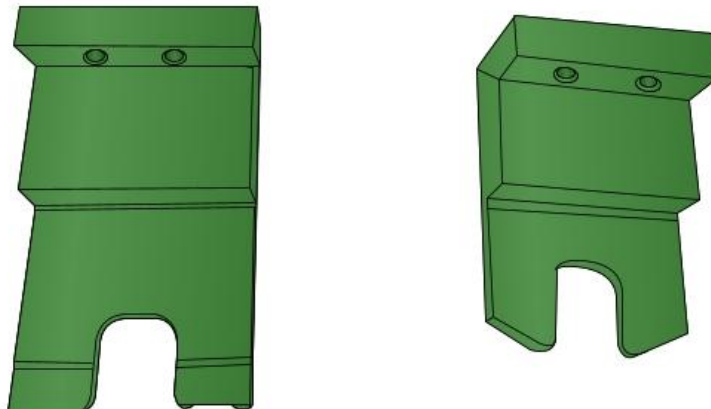


Camshaft locking tool: (2 456 372/1)

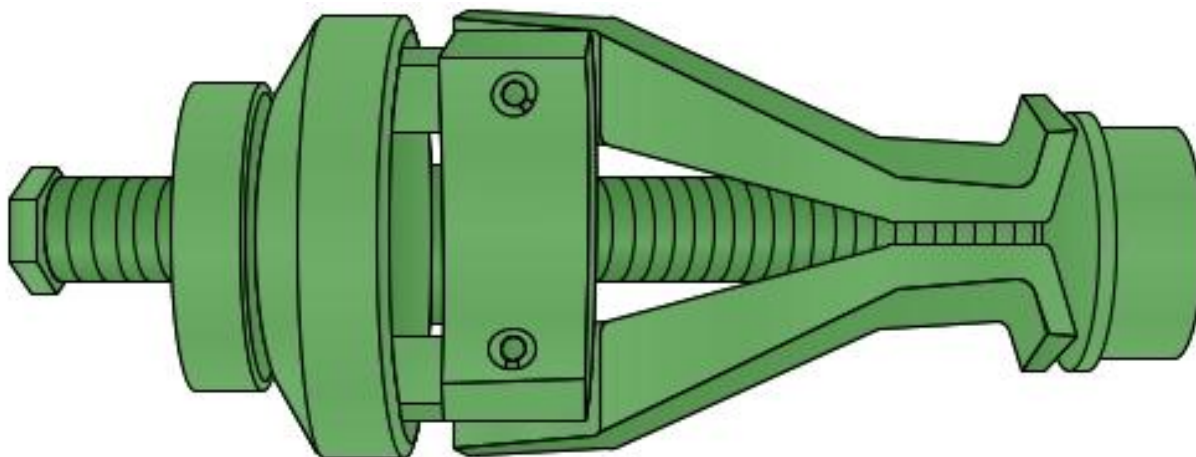


Camshaft blocking tool: (2 358 122/1)

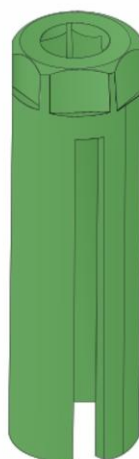
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Oil seal removal tool: (23 0 120)

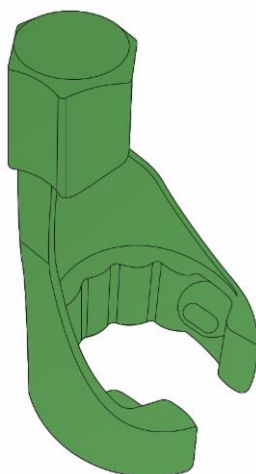


Socket wrench: (11 7 020)

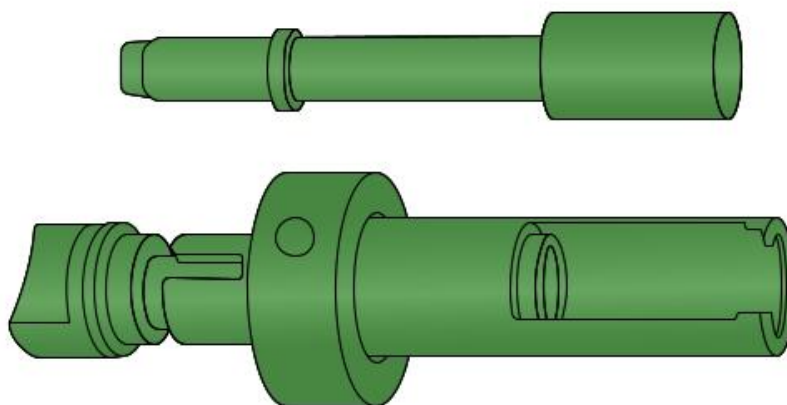


Special tool: (5 A19 557)

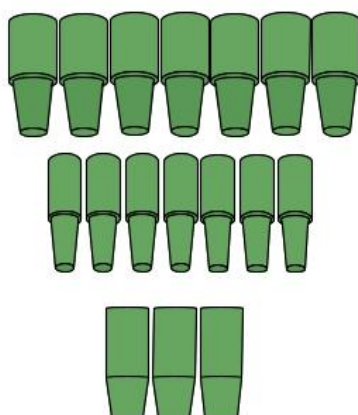
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Blanking plug: (13 5 161, 13 5 162)



Blanking plugs: (2 364 711)



Socket wrench: (2 450 487)

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Ratchet head: (0 496 855)

